

TIC 333145823

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	0.898458 d	VSX	no match
Epoch (t0)	1468.9869 BTJD	ASAS-SN	no match
Duration	0.0400 d	Gaia eclipsing-binary table	no match
Depth	1.183 % (11,830 ppm)	Gaia variable classification	no match
SNR	270.0	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	28 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	1.18 (gate >= 0.50)		
Robust depth sigma	113.2 (gate >= 7.0)		
Odd/even z-score	0.49 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 33 light-curve dip in this sector: 70.5 sigma t0 refined by -0.5000 phase

Difference-image peak SNR 9.47 (gate >= 5.0) Compactness 0.90 (gate >= 0.50)

Dimming centroid offset 0.45 px (9 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 333145798

2.96 px away, dTmag +4.67 - farther from the dimming than the target

